

Credit EDA Case Study

Credit Risk Analysis By

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Introduction

Credit Risk Analysis plays a major part in providing the data about the repayment abilities of the loaner. In this case study, by applying Exploration Data Analysis, we will be trying to provide the basic insights of risk Analysis in Finance & Banking sectors.

This study aims on using EDA in the real time, where we will be picking up the driving factors which have higher effect in credit risk analysis and we will be showcasing how data is used in identifying the possible defaulters and thus reduce the risk of losing money while lending to the customers.

Approach

- Analyzing the problem statement
- Data Sourcing
- Analyzing the data that is useful for analysis
- Cleaning the data and handling missing values
- Plotting the data
- Univariate and Bivariate analysis using plots
- Finding top correlation
- Merging the datasets
- Getting a conclusion with the help of Univariate, Bivariate and multivariate analysis

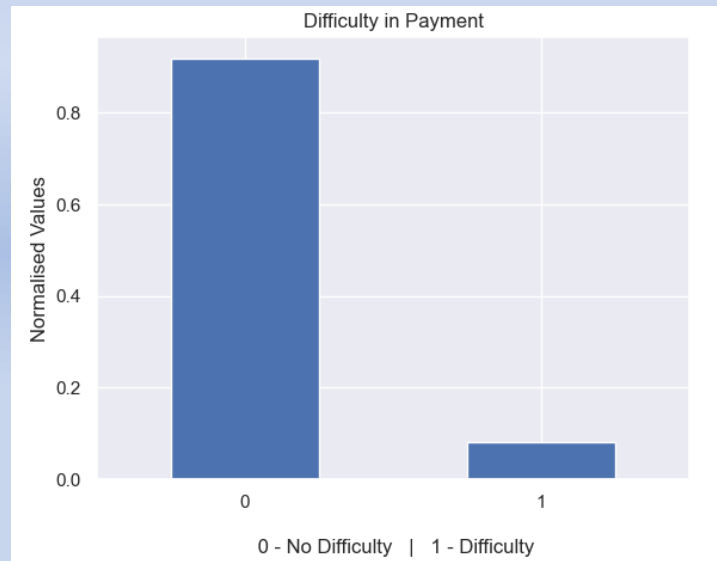
Missing Values

- For columns with more than 50% missing values the columns were dropped
- For categorical columns in most of the cases mode was filled in place of missing values
- For occupation the missing value was filled with unknown as the mode was laborers and just on the basis of mode filling the data wasn't appropriate as most of the Nan's had high salary which could mean that the missing value maybe managers or high skilled workers who don't want to disclose their occupation
- For numerical data if the data followed normal distribution mean was filled and if not mode was filled in the missing values.

EDA Insights

Univariate data interpretation on Credit Applications:

From the ratio of people having no payment difficulty to people having payment difficulty it can be inferred that 1 person out of 11 people who have received credit face difficulty in paying. i.e., there is a trend of 8% of defaulters among the applicants.



Defaulters Vs Non-Defaulters: Categorical Data Interpretation

1. When compared to revolving loans, a huge part of company's portfolio flows as cash loans comprising of 85% for Non-Defaulters and almost 95% for Defaulters.
2. Women Lead as higher Loanee with 66%. We can see that in defaulters ratio women prevails higher as near as 56%. But when compared gender wise only 2% women loanee are defaulters whereas 12% of men are defaulters.
3. 66% non car owners have taken a loan. In defaulters, the percentage is greater than 75% for people owning a car and greater than 30% for people not owning car.
4. We can see that more risk is involved in loaning to applicant who do not own a car.
5. People owning real estate take more loans and nearly 70% of applicants owning real estate are not defaulters and nearly 30% applicants not owning real estate are not defaulters.
6. 81% applicants came accompanied for loan application. But this parameter doesn't play a role in influencing payment default.

7. 52% of loan applicants are working and contribute to highest percentage among defaulters. Also there is a moderate number of pensioners with least contributing to defaulters percentage as well showing a lesser risk value.

8. Among the loanee, 71% of loan applicants have only secondary/secondary special education. The number of applicants with academic degree is minute.

9. 64% of loan applicants are married . Among defaults, married people stays high with 60% contribution.

10. 89% of loan applicants have a house/apartment.

11. 31% have no occupation mentioned. Leaving unknown the next highest applicant occupation is laborer.

12. People who work for business entity type #3 take more loans.

13. we are able to see that maximum number of applicants falls in the age range of 35-45.

14. Even though Maximum number of applicants are from Medium income group, we can see that loan request from very high and very low income groups are comparatively low.

Outliers found using Scatterplot and Its interpretation:

- Found below mentioned major datasets having Outliners(Note: Not all the plots are useful*):
"CNT_CHILDREN", "DAYS_REGISTRATION","DAYS_ID_PUBLISH", "CNT_FAM_MEMBERS", 'AMT_INCOME_TOTAL',
'AMT_CREDIT', 'AMT_ANNUITY', 'AMT_GOODS_PRICE', DAYS_LAST_PHONE_CHANGE.

Findings:

1. The density plot of SK_ID_CURR have multimodal distribution
2. The number of outliers seem to be more for continuous data
3. AMT_CREDIT is low for defaulters which is good for the company
4. AGE_IN_YEARS column has density to be more at 30 which means more defaulters are found around this age
5. EXT_SOURCE_2 shows higher density at Target 0
6. OBS_30_CNT_SOCIAL_CIRCLE, DEF_30_CNT_SOCIAL_CIRCLE, OBS_60_CNT_SOCIAL_CIRCLE, DEF_60_CNT_SOCIAL_CIRCLE shows high Target 1
7. More number of defaulters seem to have changed phone earlier when compare to non-defaulters which may indicate will issue in repaying
8. AMT_REQ_CREDIT_BUREAU_YEAR, month, week, hour shows that Target 1 people want to take loans from various companies.
9. REGION_RATING_CLIENT' & 'REGION_RATING_CLIENT_W_CITY' -the highest loan applicants are found in region 2

Bivariate Analysis:

1. Even though the highest number of loans are taken by medium income people, the highest number of defaulters are found in High Income people as the AMT_CREDIT is also high.
2. Male applicants are lesser but still the ratio of defaulters is more for male applicants.
3. Married people in the age group of 25-35 and 35-45 have difficulty in paying back

Interpretation from Merged Data of current and previous application

1. In the previous dataset most of the loans are consumer loans:

Approved loans are 62% and it can be seen that the data is imbalanced & 73% are repeaters.

2. 7.5% of defaulters are there in approved loans.

3. The applicants who had got their application refused, cancelled and unused have also got loan and now are facing difficulty with paying it back.

4. Working professionals whose loan status is approved have the highest number of defaulters

5. It is a notable factor that higher credit is offered to maternity leave and unemployed applicants.

6. In unused offer, the credit is small and mostly why the people are not using it.

Future Scope

- Customizing the rules based on the income because high income applicants might tend to take high loans and if not returned can cause huge loss to company
- Identifying new variables or using the existing to predict a pattern
- Using a model to predict the outcome by training it
- Evaluating the model continuous to check for improvements
- Get insights from customers and implement it to modify the model

Conclusion & Recommendation

From the interpretations made out of this EDA, We have achieved the goal of finding the high risk factors that determines the credibility of an applicant. With above findings, the following recommendations are done for achieving minimal loss for the firm while lending the money. This Study also shows the set of vulnerable group that can be avoided.

Recommended Applicants:

1. Employed Females are comparatively favorable than male
2. Educated clients with Higher- Medium income.
3. Clients who got their previous applications approved.
4. Clients falling under type 3 business entity with House.
5. Senior citizens in all categories.
6. Married clients compared to other family status.
7. State servants & High skilled Technical staffs.

High risk Applicants:

1. Previously refused, cancelled or unused offer clients.
2. Poor external source scorer.
3. Unemployed clients.
4. Low age groups are more vulnerable than mid age applicants.
5. Applicants not having any assets and falls under low income.

Thank you